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16 February 1988
SUPERSEDING
MIL-A-55059C
3 November 1975

MILITARY SPECIFICATION

ADAPTER, BATTERY TERMINAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers adapters for attaching ring lugs on cables to the positive and negative storage battery terminal posts (see 6.1).

1.2 Classification. Terminal adapters shall be of the following types, as specified (see 6.2):

Type I	- Positive
Type II	- Negative

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5940
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SPECIFICATIONS
FEDERAL

QQ-S-698	- Steel, Sheet and Strip, Low Carbon.

MILITARY

MIL-L-13808	- Lead Plating (Electrodeposited).

STANDARDS
FEDERAL

FED-STD-H28/2	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-45662	- Calibration Systems Requirements.
MS75004	- Adapter, Battery Terminal (Storage Battery).

(Copies of specifications and standards required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish terminal adapters, of the specified type (see 6.2), which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the units to be furnished to the Government. All subsequent terminal adapters delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in applicable specifications or standards. Materials shall be free of defects which adversely affect performance or serviceability of the finished product (see 4.7.1).

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3.2.1 Body. The body of the terminal adapter shall be a die cast from a lead-antimony alloy having 4 to 7 1/2 percent (%) antimony, not more than 1% materials which are neither lead nor antimony, and the remainder lead (see 4.7.1).

3.2.2 Insert. The insert used in the terminal adapter body shall be low carbon steel conforming to QQ-S-698 and shall be either (a) hot rolled (HR), sheet or strip, pickled and oiled (PO) or (b) cold rolled (CR), commercial quality (temper 2-5) (see 4.7.1).

3.2.3 Bolts and nuts. The bolts and nuts used for clamping to the battery terminal post and for the mounting of leads shall be steel having a tensile strength of not less than 55,000 pounds per square inch (psi) (see 4.7.1).

3.2.4 Threaded parts. The screw threads of the form, number per inch and class specified on the applicable military standard shall be in accordance with FED-STD-H28/2 (see 4.7.1).

3.2.5 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.3.1).

3.3 Design and construction. Terminal adapters shall be constructed with a die cast body having a steel insert for reinforcing the core and provisions for a bolt and a nut for clamping to the battery posts and a bolt and nut for mounting the leads. The insert shall be imbedded in and completely covered by the body alloy and all dimensions shall be in accordance with MS75004 as specified below (see 4.7.2):

Type	Part number
I	MS75004-1
II	MS75004-2

3.4 Finish. Bolts and nuts shall be lead plated in accordance with type II, class 2 of MIL-L-13808 (see 4.7).

3.5 Assembly. Completed terminal adapters shall have bolts and nuts assembled in place (see 4.7.2).

3.6 Terminal identification. The letters "P" for type I or "N" for type II and the signs "+" for type I or "-" for type II shall be cast, engraved or stamped on the top face of each terminal adapter for identification of positive or negative terminals as applicable (see 4.7.2).

3.7 Marking. The terminal adapters shall be marked in accordance with MIL-STD-130 and shall consist of the MS part number and manufacturers identification (see 4.7.2).

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3.8 Workmanship. The workmanship shall be such as to insure a product free of sharp edges. The insert shall be completely covered by the lead alloy and die cast body shall be free of all voids or pits which would affect its serviceability (see 4.7.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 $\pm$ 18°F
b. Barometric pressure $28.5 \begin{smallmatrix} +2 \\ -3 \end{smallmatrix}$ inch (in) mercury (Hg)
c. Relative humidity 50 $\pm$ 30%

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4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select two terminal adapters, of the specified type, produced under the production contract for first article inspection. First article samples shall be inspected as specified in table I and table II. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply terminal adapters that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	QCI 1/
Materials and construction	3.2 thru 3.2.5	4.7.1	X	X
Defects (see table II)	3.3 and 3.5 thru 3.8	4.7.2	X	X

1/ Quality Conformance Inspection.

4.4.1 First article inspection failure. Item deficiencies determined during, or as a result of, the first article inspection shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, the first article inspection shall be evidence that all items already produced prior to completion of the first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article inspection is completed to the satisfaction of the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. A lot shall consist of all the terminal adapters of one type, from an identifiable production period, offered for delivery at one time.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL).

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Classification	AQL
Major	2.5
Minor	4.0

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 2.5% Defective	
101	Incorrect dimensions affecting Interchangeability not within tolerance (see 3.3).	SIE 1/ SIE
102	Incomplete assembly (see 3.5).	Visual
103	Improper terminal identification (see 3.6).	Visual
Minor	AQL 4.0% Defective	
201	Incorrect dimensions not affecting interchangeability (see 3.3).	SIE
202	Improper marking (see 3.7).	Visual
203	Faulty workmanship (see 3.8).	Visual

1/ SIE = Standard Inspection Equipment.

4.6 Failure. Failure of any terminal adapter to pass any of the specified quality conformance inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.7 Methods of inspection.

4.7.1 Materials and construction. Conformance to 3.2 through 3.2.5 and 3.4 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.7.2 Defects. Conformance to 3.3 and 3.5 through 3.8 shall be determined by examination for the defects listed in table II. Examination shall be visual or by measurement with standard inspection equipment.

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5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The terminal adapters furnished under this specification are used for mounting and fastening positive or negative electrical leads to SAE tapered battery terminals utilizing the nuts and bolts provided within the terminal adapters.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of terminal adapter required (see 1.2).
c. If first article inspection is not required (see 3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If first article sample size shall be other than as specified (see 4.4).
h. Selection of applicable level and packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.2).

6.3.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.4 Subject term (key word) listing.

Battery terminal, adapter
Terminal adapter, battery

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6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AT
Navy - MC
Air Force - 85

Review activities:
Navy - AS
Air Force - 99
DLA - GS

User activities
Army - ER, ME
Navy - YD

Preparing activity:
Army - AT

(Project 5940-1010)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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